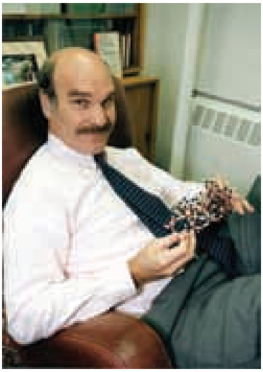

Father of molecular-scale electronics

formulated long before the means existed to test it, was so radical that it wasn't pursued or even understood for another 15 years. For this ground-breaking work, Ratner is credited as the "father of molecular-scale electronics" and his contributions were recognised in 2001 with the Feynman Prize in Nanotechnology.

SERS (1977)

In 1977 Richard P. Van Duyne from the Northwestern University discovered Surface Enhanced Raman Spectroscopy (SERS), which he built upon C.V. Raman's discovery that the scattering of light by molecules could be used to provide information about a sample's chemical composition and molecular structure. The discovery of SERS completely transformed Raman Spectroscopy from one of the least sensitive to one of the most sensitive techniques in all of molecular spectroscopy. Today, SERS is used to study the chemical reactions of molecules in electrochemistry, catalysis, materials synthesis, and biochemistry.

Scanning Tunneling Microscope (1981)

A STM in a Modern Lab

In 1981, the scanning tunneling microscope (STM) was invented by Gerd Binnig and Heinrich Rohrer at IBM's Research Laboratory in Zurich, Switzerland. This invention allowed scientists not only to observe nanoscale particles, atoms, and small molecules, but to control them. The STM helps researchers determine the size and form of molecules, observe defects and abnormalities, and discover how chemicals interact with the sample. Binnig and Rohrer won the Nobel Prize for this discovery in 1986, which quickly became standard equipment in laboratories throughout the world.

BuckyBall (1986)

Another nanotechnology breakthrough occurred in1985, when Richard Smalley, Robert Curl and graduate student James Heath at Rice University,